
CONTEXTUAL LEARNING FOR INTERRUPTION TOLERANCE AND MULTITASKING

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ABSTRACT

This project is an effort to combine the Morphognosis and Mona neural network models into a comprehensive model for learning and behavior called Mandala. Mona features a contextual causation learning with goal-directed motivation. Morphognosis features contextual multilayer perceptron (MLP) learning. Mandala achieves this by externally accumulating tiers of temporal states that are fed into an MLP at each time step. Natural environments abound in event streams that require interruption tolerance and multitasking. Mandala affords multitasking as it is robust in the presence of intervening events representing overlaid causation streams, a capability that conventional recurrent artificial neural networks (RNNs) struggle with. The results bear this out. In addition, externally accumulating temporal information discretely labels hierarchical cause-and-effect relationships that can be used for augmented processing. In the case of Mona, channeling motivation through the network for the purpose of goal-seeking requires this feature.

Keywords: Mona, Morphognosis, causation learning, multitasking, artificial neural network, machine learning.

INTRODUCTION

Hierarchical Reinforcement Learning: Assignment of Behaviours to Subpolicies by Self-Organization

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<https://www.geeksforgeeks.org/deep-learning/sparse-autoencoders-in-deep-learning/>

Remember to check EverNote and Melendey for refs.

Mona:

A pair of cooperating nest-building and foraging birds.

See: <http://tom.portegys.com/research.html#nestingbirds>

Morphognosis:

Honey bees forage for flower nectar cooperatively.

See: http://tom.portegys.com/research.html#honey_bees

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The design has context tier 0 features being assigned to sensory events that are significant to learning and recognizing pattern hierarchies. This assignment, as well as the higher tier context processing, takes place outside of the neural network. So the system as a whole consists of both the neural network and the process that accompanies the network. In a real-world scenario, significant sensory events might be determined by attention and/or emotional impact.

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Other recurrent architectures with external memory. Also rely on RNN/LSTM though. Not as integrated as Mandala.

AI Overview

Recurrent Neural Networks (RNNs) with external temporal memory are

advanced architectures designed to overcome the limitations of standard RNNs (like LSTMs or GRUs) in memorizing long-term dependencies. While standard RNNs rely on internal hidden states that often suffer from vanishing gradients over long sequences, these models integrate a separate, addressable memory bank to store and retrieve past information over extended time horizons.

Core Components and Functionality

Controller (RNN): A standard neural network (e.g., LSTM) that processes inputs and decides, via a controller, what information to write to or read from the external memory.

External Memory Bank: A matrix or tensor (often addressable) that acts as a long-term storage, allowing the model to "recall" information from far in the past.

Addressing Mechanism: Techniques like content-based addressing or location-based addressing allow the model to select specific memory locations to read from or write to.

Key Architectures

Neural Turing Machines (NTMs): A foundational model that couples a neural network controller to an external memory bank, capable of learning to manipulate and manage the memory to solve tasks like sorting or copying.

Memory Networks (MemNN/MemN2N): These models use external memory to store input data, which can be queried later to aid in tasks like question answering.

External Addressable Long-Term and Working Memory (EALWM)-Augmented RNNs: An advanced structure that splits external memory into distinct long-term and working memory parts, improving the ability to learn long-term dependencies without gradient issues.

Differentiable Neural Computers (DNCs): A further extension of NTMs that provides better dynamic memory allocation and temporal memory linkage.

Advantages

Long-Term Dependencies: Capable of capturing relationships over thousands of steps, whereas LSTMs may fail.

Improved Memory Capacity: The memory capacity is decoupled from the number of hidden units in the controller, allowing for much larger, efficient storage.

Task Versatility: Highly effective for complex tasks such as algorithmic learning, language modeling, question answering, and reasoning, which require tracking historical data.

Applications and Performance

These models are specifically used in scenarios demanding significant context, such as:

Language Understanding: Associating tags to words in long text sequences.

Algorithm Learning: Tasks like copying, sorting, and associative recall.

Question Answering: Recalling information from stories or documents.

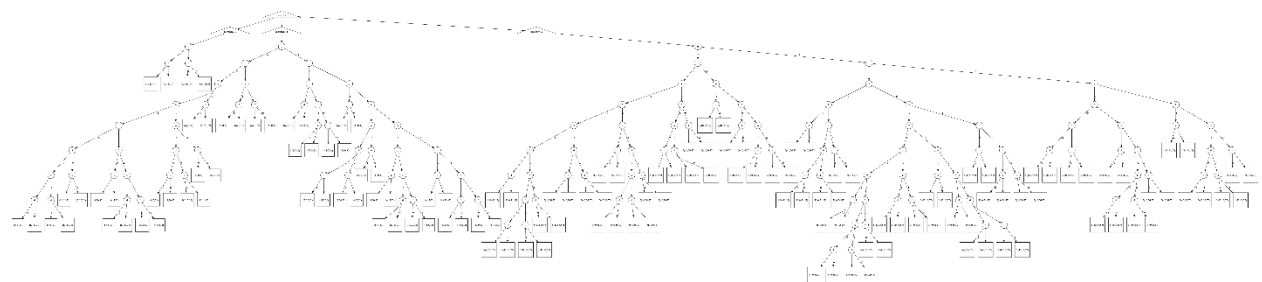
They are often compared against standard LSTM models, showing superior performance in scenarios where memory length requirement exceeds the inherent capacity of the LSTM's gating mechanism.

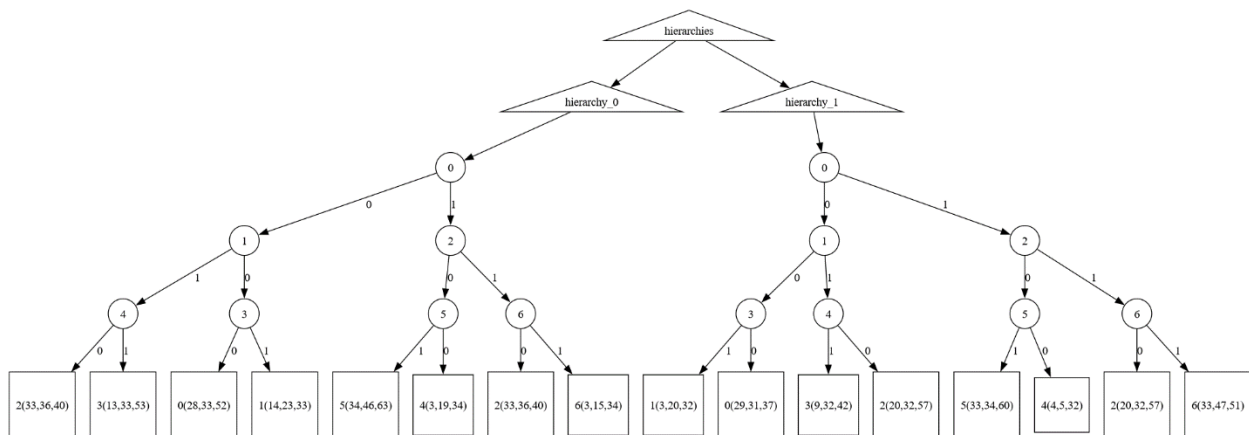
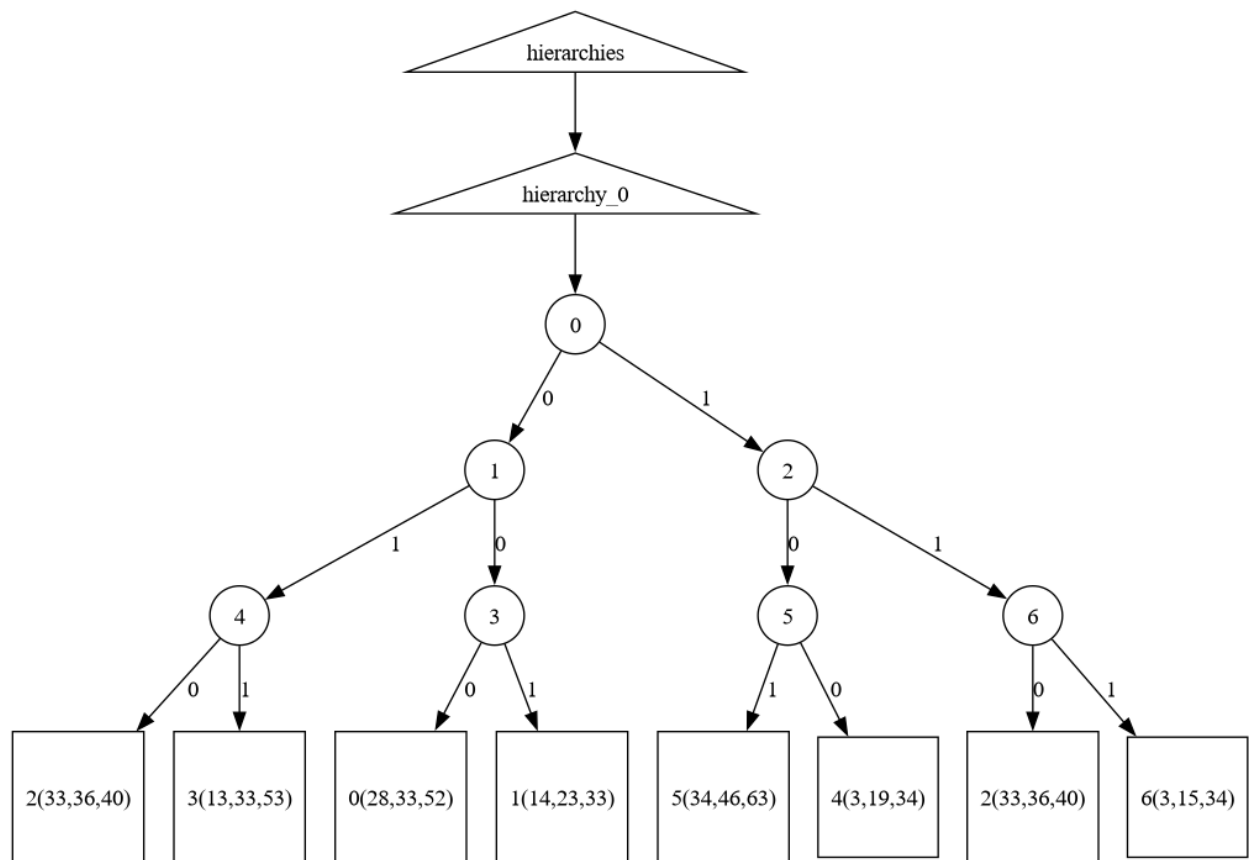
<https://arxiv.org/abs/1506.00195>

Recurrent Neural Networks with External Memory for Language Understanding

This model has a long-term memory that is addressable from the RNN and which feeds into the hidden memory.

DESCRIPTION





RESULTS

Tried Claude, Gemini, ChatGPT on test_0 sequences.

Given these sequences:

0,1,2,3

4,5,2,6

Predict the next number in these sequences:

8,8,7,0

7,11,4

7,4,5,11,11,8,10,10,10,11,2

0,1,9,11,10,2

7,9,11,9,7,8,4,5,7,10,10,8,7,2

7,0,1,11,10,7,7,10,2

4,5,10,8,8,2

11,8,0,1,8,10,2

11,7,7,11,11,0,1,11,7,11,11,9,11,10,7,2

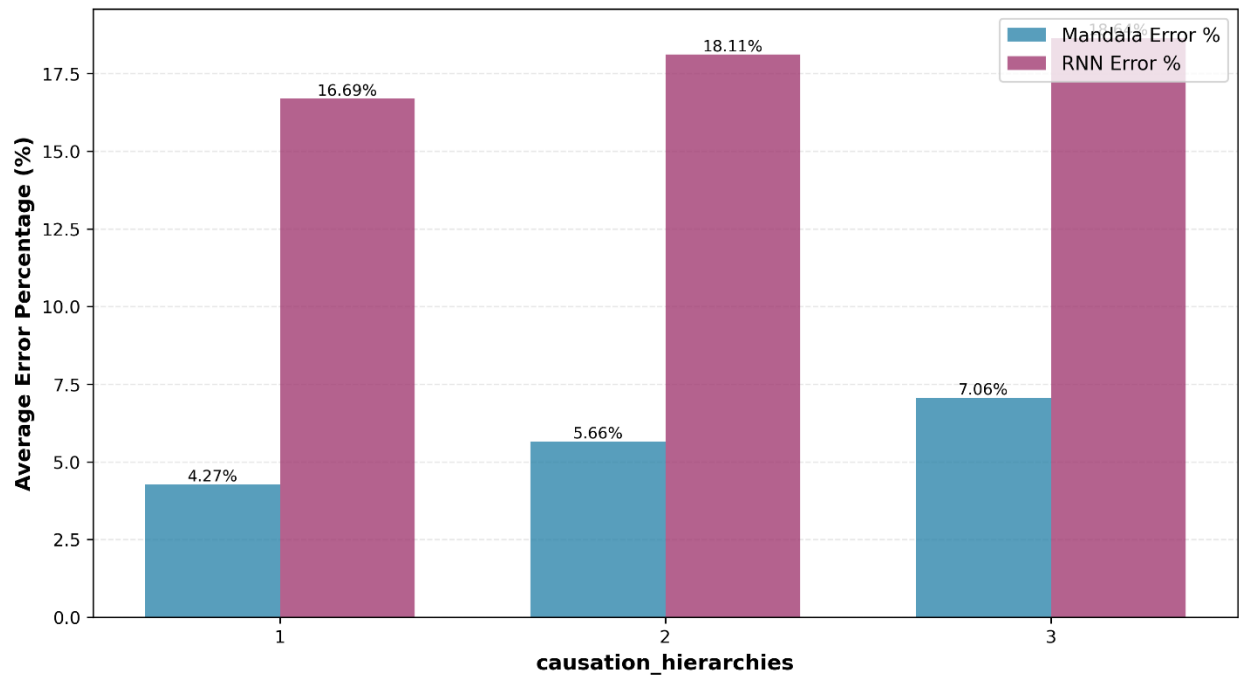
10,8,9,7,4,5,9,8,8,10,11,11,7,9,9,11,2

Note that 4,5 causes 2,3 and 0,1 causes 2,6

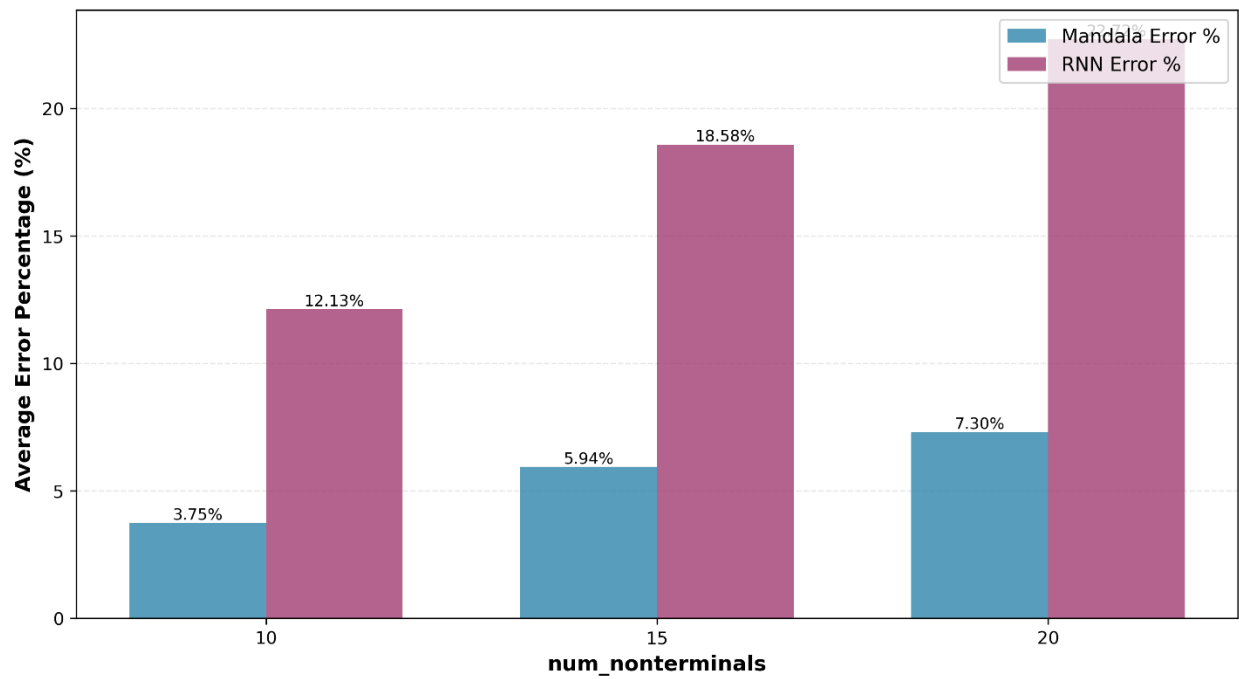
Claude predicted the first two simple sequences, But failed on the sequence that depended on recognizing the earlier transitions determine the ending transitions.

ChatGPT did not correctly predict any of the sequences.

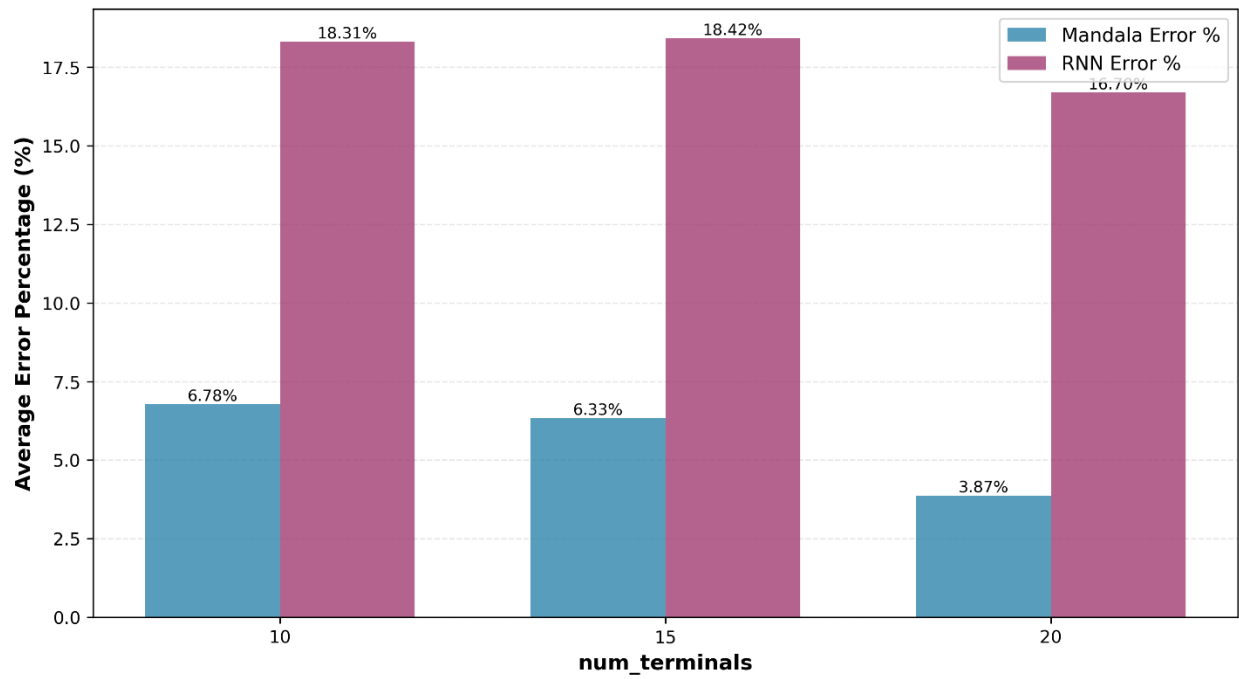
Mandala vs RNN: Average Error



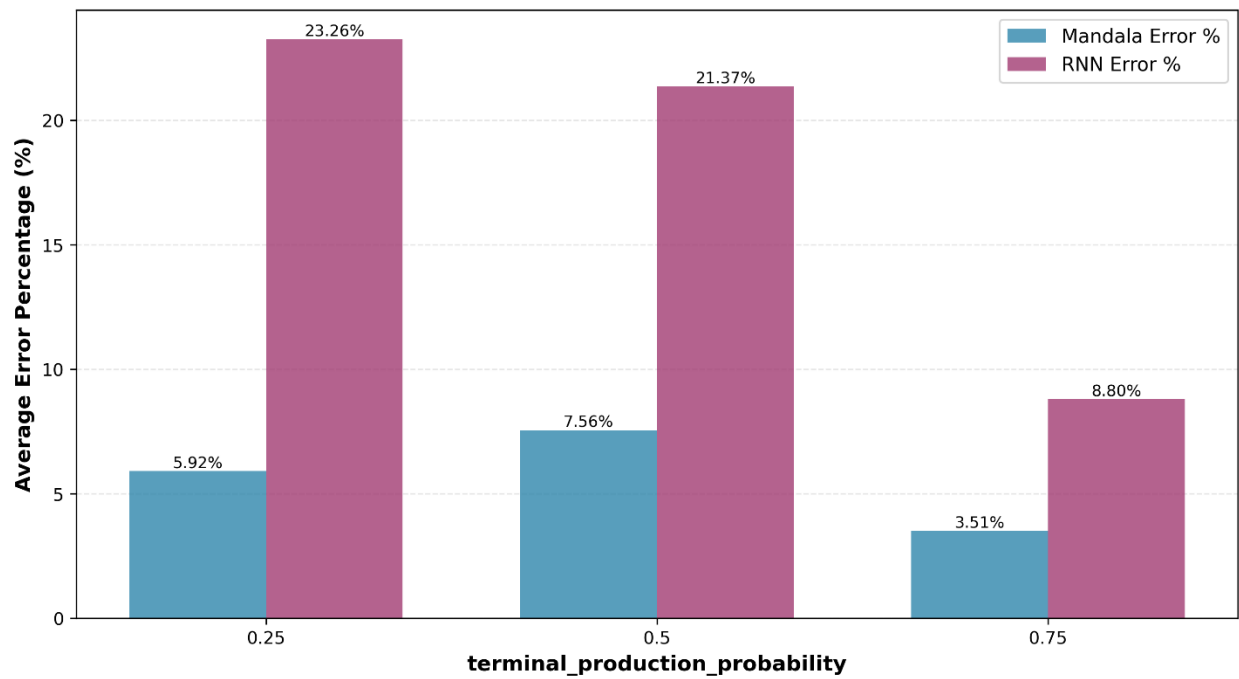
Mandala vs RNN: Average Error

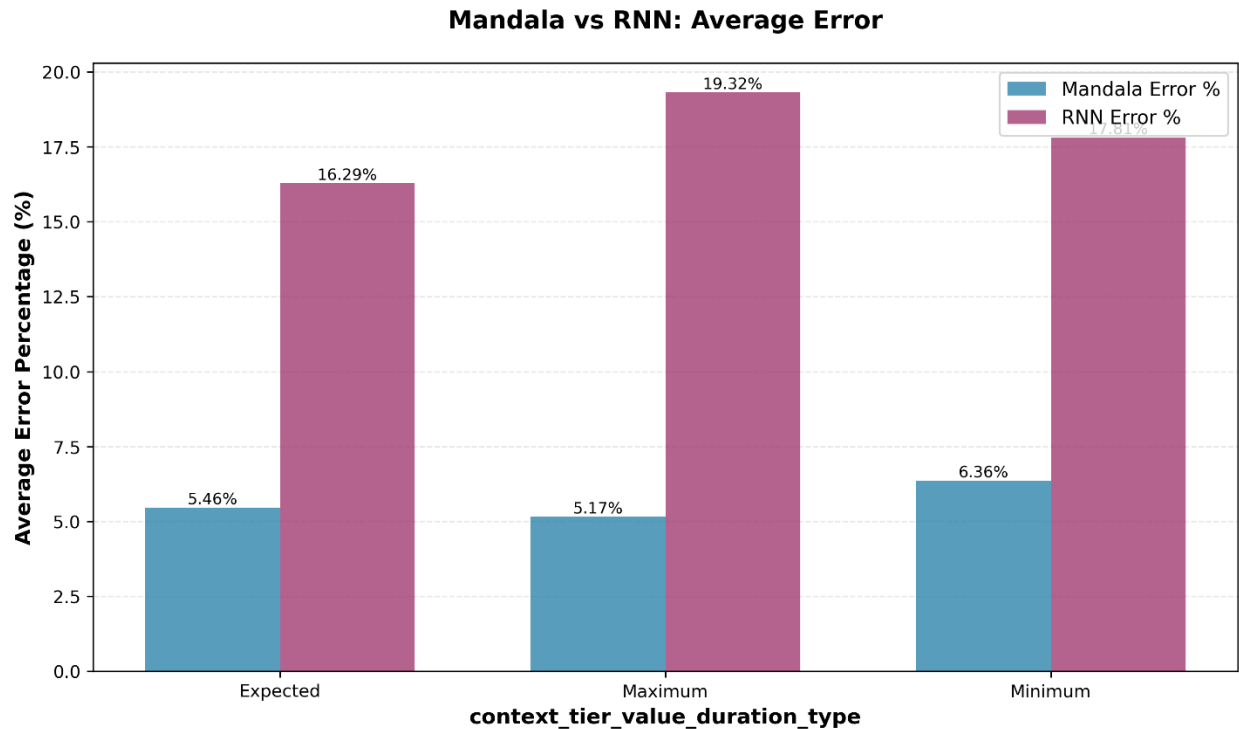


Mandala vs RNN: Average Error



Mandala vs RNN: Average Error





Generated by Claude with this prompt:

I want to create a bar chart from the included excel spreadsheet. On the X axis use the unique values from the context_tier_value_duration_type, and plot the aggregated values of mandala_error_pct and rnn_error_pct.

CONCLUSION

Note for paper: possible to learn out-of-order streams, but not done here. Reference my paper on conjunctions.

REFERENCES